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Impact of Resonant Compton Scattering on Magnetar X-Ray Polarization with Quantum Electrodynamics Vacuum Resonance

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Abstract

Recent observations have revealed significant soft X-ray polarization from several quiescent magnetars, including the intriguing 90° polarization angle (PA) swing as a function of photon energy for some sources. We thus present a general semianalytical framework for calculating energy-dependent soft X-ray polarization signatures from magnetars, consistently incorporating both quantum electrodynamics (QED) vacuum resonance in the atmosphere and resonant Compton scattering (RCS) in the magnetosphere. Starting from the polarized radiative transfer equation for RCS and treating vacuum-resonance-induced mode conversion as an input, we employ a first-order approximation in RCS optical depth to evaluate the effects of magnetospheric plasma density (which depends on magnetic twist), drift velocity, temperature, and viewing geometry on the observed radiation. Our analysis reveals that magnetic twist and plasma drift velocity are the critical parameters controlling the impact of RCS on both the absolute polarization degree and its variation across the soft X-ray spectrum. We find that sufficiently strong RCS can wash out the PA swing caused by vacuum resonance. Furthermore, in addition to the QED vacuum resonance effect, significant relativistic signatures arising from plasma drift velocity ($\beta_0 \gtrsim 0.5$) may introduce an extra 90° PA swing in the spectrum. Our calculation framework, based on the single-scattering approximation, bypasses the need for complex, multidimensional Monte Carlo simulations, providing an analytical pathway for modeling full-surface emission and rotational-phase-resolved radiation from magnetic neutron stars in support of current and future X-ray polarization missions.

Unified Astronomy Thesaurus concepts: Neutron stars (1108); Magnetars (992); Magnetic fields (994); X-ray astronomy (1810); Radiative transfer (1335); Spectropolarimetry (1973)

1. Introduction

Magnetars are isolated neutron stars (NSs) endowed with extremely strong magnetic fields (R. C. Duncan & C. Thompson 1992; C. Thompson & R. C. Duncan 1995; C. Kouveliotou et al. 1998; V. M. Kaspi & A. M. Beloborodov 2017). Their surface dipole magnetic fields inferred from spin-down measurements typically reach $B_{\rm dip} \sim 10^{14}$–10^{15} G. Historically, the magnetar subclasses of soft gamma repeaters (SGRs) and anomalous X-ray pulsars (AXPs) were distinguished by their burst behaviors, but both emit persistent soft X-rays (0.5–10 keV) in quiescence with typical luminosities of $L_{\rm X} \sim 10^{33}$–10^{36} ergs^{-1}, much larger than their spin-down luminosities. Their quiescent X-ray spectra are commonly described by a combination of a thermal blackbody with $kT \sim 0.5$–1 keV and a nonthermal power law with photon index $\Gamma \sim 2$–4 (P. M. Woods & C. Thompson 2006). They also emit hard X-rays extending to ∼100 keV with no evidence of a spectral break below 70–100 keV, indicating an active corona surrounding magnetars (V. M. Kaspi & A. M. Beloborodov 2017). Despite the rich phenomenology, the integrated X-ray spectrum alone is often inadequate to disentangle the degeneracy among different emission mechanisms and magnetospheric physics. However, X-ray polarization measurements can help break this degeneracy and provide constraints on various physical parameters and the magnetospheric structure (R. Fernández & S. W. Davis 2011; M. C. Weisskopf et al. 2022; R. Taverna & R. Turolla 2024).

In recent years, the Imaging X-ray Polarimetry Explorer (IXPE) mission (M. C. Weisskopf et al. 2016) has enabled polarimetry measurements in the 2–8 keV band, delivering high-significance detections across multiple sources, including several magnetars. IXPE observations of magnetars have revealed diverse energy-dependent polarization behaviors in the soft X-ray band: Some (e.g., 4U 0142+61 in R. Taverna et al. 2022) exhibit a sharp ∼90° polarization angle (PA) swing at around 4–5 keV; some might show signs of two ∼90° swings (e.g., SGR 1806-20 in R. Turolla et al. 2023), although with low statistical significance; while others (e.g., 1RXS J170849.0-400910 in S. Zane et al. 2023) maintain a nearly energy-independent constant PA.

To explain these spectral polarization features from IXPE results, current theoretical efforts mainly focus on two physical mechanisms: emission from a condensed surface (R. Taverna et al. 2022; R. Taverna & R. Turolla 2024) followed by resonant Compton scattering (RCS) in the magnetosphere (R. Fernández & S. W. Davis 2011; R. Taverna et al. 2014), and the mode-conversion effect via vacuum resonance in atmospheric plasma under strong magnetic fields (D. Lai & W. C. Ho 2002, 2003a; D. Lai 2023; R. M. E. Kelly et al. 2024). A condensed surface requires sufficiently low temperature and high surface binding energy, which remain uncertain (Z. Medin & D. Lai 2006a, 2006b, 2007; A. Y. Potekhin & G. Chabrier 2013). RCS by charged particles in a twisted magnetosphere can modify the emergent radiation spectrum and polarization profile. This process has been modeled in both

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semianalytical one-dimensional frameworks (M. Lyutikov & F. P. Gavriil 2006) and multidimensional Monte Carlo simulations with detailed spectral fits (R. Fernández & C. Thompson 2007; L. Nobili et al. 2008a; N. Rea et al. 2008; S. Zane et al. 2009), and further extended to polarization predictions (R. Fernández & S. W. Davis 2011; R. Taverna et al. 2014, 2020; R. Taverna & R. Turolla 2024).

Vacuum resonance, on the other hand, arises from the "competition" between plasma-induced birefringence and vacuum birefringence due to quantum electrodynamics (QED) in strong magnetic fields. When a photon crosses the resonant layer, it may undergo adiabatic mode conversion, leading to a significant change in its polarization (D. Lai & W. C. Ho 2002, 2003a, 2003b). After leaving the atmosphere, the photon may experience significant RCS in the extended magnetosphere or corona, leading to further modifications in its polarization.

In this work, we present a general three-dimensional semianalytical framework that unifies the treatment of vacuum resonance in the atmosphere and RCS in the magnetosphere, with the goal of obtaining the soft X-ray polarization of magnetic NSs. To preserve the physical transparency and geometric modularity of our calculation without relying on full Monte Carlo simulations of radiative transfer, we use the single-scattering approximation. To focus on the key physics, we adopt simplified yet testable idealized assumptions for the polarized surface emission, including the vacuum resonance effect. The structure of this paper is as follows: Section 2 introduces the physical model and computational procedure; Section 3 presents the validation of our fiducial model; Section 4 investigates the X-ray polarization signals for various physical parameters; and Section 5 provides a summary and outlook.

2. Physical Framework

To model how RCS and QED vacuum resonance jointly shape the X-ray polarization from magnetars, we distinguish the main physical processes with their characteristic locations. This section presents our physical framework for the global magnetosphere, polarized emission and propagation, the RCS formalism, and the computational workflow.

2.1. Global Magnetosphere Model

One of the key characteristics of magnetars is the presence of a highly twisted and dynamically evolving toroidal magnetic field in their magnetospheres. This arises from magnetic stress built up in the crust, which induces instabilities, leading to displacement of surface elements and distortion of the magnetic field lines anchored to them. C. Thompson et al. (2002) presented a general self-similar, force-free, globally twisted magnetic field model that has been widely used in previous studies (e.g., R. Fernández & C. Thompson 2007; L. Nobili et al. 2008a; R. Fernández & S. W. Davis 2011).

For the purposes of this work, we adopt a simple parameterized model in which a toroidal component proportional to the poloidal field is added, i.e.,

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{B_{\rm p}}{2}\left(\frac{r}{R}\right)^{-3}(2\cos\theta,\ \sin\theta,\ \xi_\tau \sin\theta), \tag{1}$$

where $B_{\rm p}$ is the surface polar dipole field strength, R is the magnetar radius, and the parameter ξ_τ quantifies the global twist (hence the current distribution). This choice is not strictly force-free, but it captures a quasi-static configuration adequate for our semianalytic treatment.

We assume that the magnetospheric plasma consists of a single charge species and that the ions are stationary. Given $\boldsymbol{B}$, the spatial distribution of current-carrying electrons along the closed field lines measured in the stellar frame is prescribed as (C. Thompson et al. 2002; L. Nobili et al. 2008a; R. Taverna et al. 2020)

$$n_e(\boldsymbol{r},\ \beta_0) = \frac{(\xi_\tau \beta_0^{-1})}{4\pi e}\frac{B(\boldsymbol{r})}{r}, \tag{2}$$

where $B = |\boldsymbol{B}(\boldsymbol{r})|$. We note that kinetic modeling of external currents in magnetars suggests that the real magnetospheric plasma may be a denser electron–positron pair plasma with high $e^{\pm}$ multiplicity (C. Thompson 2008; A. M. Beloborodov 2013; C. Thompson & A. Kostenko 2020; J. Zhang & C. Thompson 2025; see Section 5 for further discussion). We denote by $\beta_e \in (-1, 1)$ the electron velocity (normalized by c) along the field lines, and $\beta_0 = |\langle\beta_e\rangle|$ the mean drift speed. To determine the full phase-space distribution, we assume a one-dimensional, relativistic, thermal (Maxwell–Jüttner) velocity distribution along the field line, boosted by β_0 (see also L. Nobili et al. 2008a, 2008b):

$$f_1(\beta_e)\, d\beta_e = \frac{\gamma_0\gamma_e^3(1-\beta_e\beta_0)\exp(-\gamma'/\Theta_e)}{2K_1(1/\Theta_e)}\, d\beta_e, \tag{3}$$

with $\Theta_e = kT_e/m_ec^2$, $\gamma' = \gamma_e\gamma_0(1-\beta_e\beta_0)$, and K_1 the modified Bessel function. This boosting captures the net drift from magnetic north to magnetic south. We assume that the magnetospheric temperature T_e is uniform, and treat T_e and β_0 as free parameters. Overall, our magnetosphere model is characterized by three parameters: (ξ_τ, β_0, T_e).

2.2. Seed Polarized Radiation and Propagation

2.2.1. Surface-emission Model

The bulk of the 2–8 keV photons probed by IXPE originate from the magnetar surface. To model the spectrum and polarization of the surface emission, it is necessary to construct self-consistent magnetic NS atmosphere models that take QED effects into account (e.g., W. C. G. Ho & D. Lai 2003; D. Lai & W. C. Ho 2003a; M. van Adelsberg & D. Lai 2006; N. Shabaltas & D. Lai 2012; R. Taverna et al. 2020; I. Caiazzo et al. 2022). Here we summarize the key features of this emission (see also D. Lai 2023).

X-ray photons (with $E = \hbar\omega \ll E_B = \hbar\omega_B = \hbar eB/m_ec = 1158B_{14}$ keV) propagating in magnetized NS atmospheric plasma have two polarization modes: The ordinary mode (O-mode) is mostly polarized parallel to the $\boldsymbol{k}$–$\boldsymbol{B}$ plane, while the extraordinary mode (X-mode) is perpendicular to it, where $\boldsymbol{k}$ is the photon wavevector and $\boldsymbol{B}$ is the external magnetic field. The two modes have very different absorption and scattering opacities, with $\kappa_{\rm X-mode} \sim u_e^{-1}\kappa_{\rm O-mode}$, where $u_e \equiv (\omega_B/\omega)^2 \gg 1$. Thus, the two modes have very different photospheric densities, with $\rho_{\rm X\text{-}mode} \gg \rho_{\rm O\text{-}mode}$, and the emergent radiation from the NS surface is dominated by the X-mode.

However, the effect of vacuum polarization can change the above picture significantly. In particular, the competition between QED-induced vacuum birefringence and plasma-induced birefringence gives rise to a vacuum resonance; for a photon of energy E, this occurs at the density

$$\rho_V \simeq 0.964\, Y_e^{-1} B_{14}^2 E_1^2 f^{-2}\ \mathrm{g\ cm^{-3}}, \tag{4}$$

where $Y_e = \langle Z/A \rangle$ is the electron fraction, $E_1 = E/(1\ \mathrm{keV})$, and $f = f(B)$ is a slowly varying function of B and is of order unity.

For $\rho \gtrsim \rho_V$ (where the plasma effect dominates the dielectric tensor) and $\rho \lesssim \rho_V$ (where vacuum polarization dominates), the photon modes (for typical $\theta_{\mathrm{kB}} \neq 0$) are almost linearly polarized, i.e., the linear X-mode and O-mode; at $\rho = \rho_V$, however, the normal modes become circularly polarized. The importance of this vacuum resonance lies in the mode-conversion phenomenon (D. Lai & W. C. Ho 2002, 2003a): An X-mode photon emerging from its photosphere (at $\rho_{\text{X-mode}}$) and propagating outward can convert to the O-mode at ρ_V, provided that its energy satisfies

$$E \gtrsim E_{\mathrm{ad}} = 2.52\, (f\, \tan\theta_{\mathrm{kB}})^{2/3} \left(\frac{1\ \mathrm{cm}}{H_\rho}\right)^{1/3} \mathrm{keV}, \tag{5}$$

where $H_\rho = |ds/d\ln\rho|$ is the density scale height (evaluated at $\rho = \rho_V$) along the ray. A low-energy ($E \lesssim E_{\mathrm{ad}}$) photon would not experience this conversion. In typical magnetar atmospheres, $\rho_{\text{X-mode}} \gg \rho_V$ is always satisfied. Depending on the relative locations of the O-mode photosphere and vacuum resonance, the X-ray polarization signatures are qualitatively different. Assuming a fully ionized atmosphere (with ion charge Ze and mass Am_p), D. Lai (2023) finds the ratio

$$\rho_V/\rho_{\mathrm{O-mode}} = (B/B_{\mathrm{QV}})^2, \tag{6}$$

with

$$B_{\mathrm{QV}} = 7.8 \times 10^{13} \left(\frac{\mu g_2 \cos\alpha}{Z\mathcal{G} E_1 \sin^2\theta_{\mathrm{kB}}}\right)^{1/4} \left(\frac{f}{T_6^{1/8}}\right) \mathrm{G}, \tag{7}$$

where $\mu = A/(1+Z)$ is the "molecular" weight, g_2 is the surface gravity in units of $2 \times 10^{14}\ \mathrm{cm\ s^{-2}}$, T_6 is the surface temperature in units of 10^6 K, and $\mathcal{G} = 1 - \exp(-E/kT)$. For magnetars with surface magnetic fields satisfying $B \lesssim B_{\mathrm{QV}}$, the vacuum resonance lies at a lower density than $\rho_{\mathrm{O-mode}}$; the emergent radiation is dominated by the X-mode for $E \lesssim E_{\mathrm{ad}}$ and by the O-mode for $E \gtrsim E_{\mathrm{ad}}$, i.e., there is a 90° rotation of the PA between $E \lesssim E_{\mathrm{ad}}$ and $E \gtrsim E_{\mathrm{ad}}$. On the other hand, for $B \gtrsim B_{\mathrm{QV}}$, the vacuum resonance lies between $\rho_{\text{O-mode}}$ and $\rho_{\text{X-mode}}$, and the emergent radiation from the surface is dominated by the X-mode for all values of E.

Quantitative calculations of the atmospheric emission (spectrum and polarization) have been fully described in previous works (e.g., W. C. G. Ho & D. Lai 2003; M. van Adelsberg & D. Lai 2006; D. Lai 2023). In this paper, since our focus is on the effect of magnetospheric scattering, we adopt a simple prescription for the surface emission. For a given spot (located at the polar angles θ_s, ϕ_s) with temperature T_s, the total radiative intensity (in the stellar frame) is given by

$$I_\omega^{(\mathrm{tot})}(\widehat{\boldsymbol{\Omega}}) = I_\omega^1(\widehat{\boldsymbol{\Omega}}) + I_\omega^2(\widehat{\boldsymbol{\Omega}}) \propto \frac{\mu_s^{a-1}\omega^3}{\exp(\hbar\omega/kT_s) - 1}, \tag{8}$$

where μ_s is the (stellar-frame) cosine between the emission direction ($\widehat{\boldsymbol{\Omega}}$) and the local magnetic field, and $a \geqslant 1$ is a beaming parameter (we adopt $a = 1$ for isotropic emission). As discussed above, for $B \lesssim B_{\mathrm{QV}}$, the X-ray polarization degree is set to 1 at lower energies (i.e., X-mode dominated) and to -1 at higher energies (i.e., O-mode dominated) by comparing E with E_{ad}, using a smooth transition (e.g., a Fermi–Dirac-type switch).

2.2.2. General-relativistic Effects

To describe photon propagation from the NS surface to the scattering region, general-relativistic (GR) effects must be considered. For typical magnetar spin periods $P = 2\pi/\Omega = 2$–12 s, rotational frame dragging is negligible: $(R\Omega/c)^2 \lesssim 4 \times 10^{-8} R_6^2 P_1^{-2}$ (R. Fernández & S. W. Davis 2011). Thus, the Schwarzschild metric suffices.

Key GR effects include gravitational redshift and light bending. As photons travel to $\gtrsim 10R$, where resonant scattering occurs (see below), the metric at the scattering point is almost flat, with $GM/(10Rc^2) \ll 1$. For redshift, since the photon occupation number $\mathcal{N}(\omega) \propto I_\omega/\omega^3$ is Lorentz invariant, under the frequency transformation $\omega \to g\omega$ where $g = [1 - 2GM/(Rc^2)]^{1/2}$, the intensity transforms as $I_\omega \to g^3 I_\omega$. We treat the light-bending effect using the mapping between local emission angle α and asymptotic angle ψ as in A. M. Beloborodov (2002), which is accurate for $R \gtrsim 4GM/c^2$.

2.3. Resonant Compton Scattering Formalism

In the outer magnetosphere, photons are scattered by electrons mainly through RCS. For a magnetar with dipole field B_{p} and twist parameter ξ_τ (see Equation (1)), RCS occurs at the radius (neglecting the motion of electrons; see below)

$$r_{\mathrm{sc}} = 10.5\, R \left(\frac{B_{\mathrm{p}}}{10^{14}\ \mathrm{G}}\right)^{1/3} \left(\frac{E}{1\ \mathrm{keV}}\right)^{-1/3} F(\theta), \tag{9}$$

where $F(\theta) = \left(1 - \frac{3}{4}\sin^2\theta\right)^{1/6}$. Given that the Landau-level lifetime of gyrating electrons is much shorter than the cyclotron period (R. Fernández & S. W. Davis 2011), electrons can be assumed to remain in the ground state, rendering individual scattering events effectively elastic in the electron rest frame (ERF). We neglect nonresonant scattering and recoil in this work.

2.3.1. Cross Sections

In the Thomson limit, the nonrelativistic RCS cross sections were first derived by J. Ventura (1979; see also Chapter 4 of P. Meszaros 1992). We denote the unit vector along the propagation direction by $\widehat{\boldsymbol{\Omega}}$, and the polarization states by $\alpha \in \{1, 2\} = \{\mathrm{O}, \mathrm{X}\}$. In the ERF, the differential cross section takes the form (e.g., L. Nobili et al. 2008a)

$$\left(\frac{d\sigma}{d\Omega}\right)^{(\mathrm{ERF})}_{\alpha\hat{\boldsymbol{\Omega}}' \to \beta\hat{\boldsymbol{\Omega}}} = \left(\frac{e^2}{m_e c}\right) K'_{\alpha\beta}\, \delta(\omega - \omega_B), \tag{10}$$

where $\omega = E/\hbar$ is the photon frequency, $\omega_B = eB/m_e c$ is the electron cyclotron frequency, and the coefficients $K'_{\alpha\beta}$ are

$$K'_{11} = \frac{3\pi}{8}(\widehat{\boldsymbol{\Omega}}' \cdot \widehat{\boldsymbol{B}})^2(\widehat{\boldsymbol{\Omega}} \cdot \widehat{\boldsymbol{B}})^2,$$
$$K'_{12} = \frac{3\pi}{8}(\widehat{\boldsymbol{\Omega}}' \cdot \widehat{\boldsymbol{B}})^2,$$
$$K'_{21} = \frac{3\pi}{8}(\widehat{\boldsymbol{\Omega}} \cdot \widehat{\boldsymbol{B}})^2,$$
$$K'_{22} = \frac{3\pi}{8}. \tag{11}$$

Note that all coefficients involving dot products of unit vectors should be calculated in the ERF, so they are all related to the variable β_e. When transforming to the stellar frame, relativistic beaming modifies the differential cross section:

$$\left(\frac{d\sigma}{d\Omega}\right)^{(\rm NS)} = \frac{1}{\gamma_e^2(1-\beta_e\mu)^2}\left(\frac{d\sigma}{d\Omega}\right)^{(\rm ERF)}, \tag{12}$$

where $\mu = \widehat{\boldsymbol{\Omega}} \cdot \widehat{\boldsymbol{B}}$ is measured in the NS frame, and the delta function must also transform

$$\delta(\omega - \omega_B)^{(\rm ERF)} \rightarrow \frac{1}{\gamma_e(1-\beta_e\mu)}\delta(\omega - \omega_D)^{(\rm NS)}, \tag{13}$$

with $\omega_D = \omega_B/[\gamma_e(1-\beta_e\mu)]$. We denote the differential cross section in the stellar frame as

$$\left(\frac{d\sigma}{d\Omega}\right)^{(\rm NS)}_{\alpha\hat{\boldsymbol{\Omega}}'\rightarrow\beta\hat{\boldsymbol{\Omega}}} = \left(\frac{e^2}{m_e c}\right)K_{\alpha\beta}\,\delta(\omega-\omega_D), \tag{14}$$

with $K_{\alpha\beta} \equiv K_{\alpha\beta}(\widehat{\boldsymbol{\Omega}}' \rightarrow \widehat{\boldsymbol{\Omega}}) = K'_{\alpha\beta}/[\gamma_e(1-\beta_e\mu)]^3$ the nondimensional scattering coefficients in the NS frame. The scattering radius for a photon of frequency ω can be determined by solving the resonance condition $\omega = \omega_D$ for a given emission direction and magnetospheric parameters. For the nonrelativistic cases with a dipole field, solving $\omega = \omega_D \simeq \omega_B$ yields Equation (9).

In the ERF, integrating Equation (10) over the scattered directions and summing over β gives the total cross section $\sigma_\alpha(\widehat{\boldsymbol{\Omega}}, \beta_e)$ for photons of mode α propagating along $\widehat{\boldsymbol{\Omega}}$, which is also Lorentz invariant:

$$\sigma_1 = \frac{2\pi^2 e^2}{m_e c}(\hat{\boldsymbol{\Omega}} \cdot \hat{B})^2\delta(\omega-\omega_B),$$
$$\sigma_2 = \frac{2\pi^2 e^2}{m_e c}\,\delta(\omega-\omega_B). \tag{15}$$

Using the estimate $\sigma \simeq (\pi^2 e^2/m_e c)\,\delta(\omega-\omega_B)$ and Equation (2) for n_e, we obtain the characteristic optical depth across the resonant layer (see also C. Thompson et al. 2002):

$$\tau_0(\hat{\boldsymbol{\Omega}}) \equiv \int ds\, n_e \sigma \simeq \frac{\pi\xi_\tau}{4\beta_0}\frac{B}{r_{\rm sc}}\left|\frac{dB}{ds}\right|^{-1} \sim \frac{\pi\xi_\tau}{4\beta_0}, \tag{16}$$

where ds is integrated across the RCS layer along $\widehat{\boldsymbol{\Omega}}$ and $|dB/ds| = (\hat{\boldsymbol{\Omega}} \cdot \boldsymbol{\nabla})B \sim B/r_{\rm sc}$ is the directional derivative of B along the photon path. We note that the optical depth is independent of energy. For a specific polarization mode α, the corresponding optical depth is

$$\tau_\alpha(\hat{\boldsymbol{\Omega}}) \equiv \int ds\, d\beta_e\, n_e f_1(\beta_e)(1-\beta_e\mu)\sigma_\alpha, , \tag{17}$$

where the factor $(1-\beta_e\mu)$ reflects the encounter rate between photons and electrons due to their relative motion in the NS frame. Note that $\tau_\alpha \sim 1$ indicates that the expectation value of the number of scatterings over the photon ensemble is unity in the magnetosphere.

2.3.2. Polarized Radiative Transfer in the Magnetosphere

After a photon leaves the NS surface, it propagates in the magnetosphere, whose dielectric property is dominated by vacuum polarization (J. S. Heyl et al. 2003). At small radii, the photon's polarization state evolves adiabatically following the varying magnetic field it experiences, such that the specific mode intensity is constant (except for the GR corrections discussed in Section 2.2.2).[4]

Near the RCS layer, the transfer equation for polarized intensity $I^\alpha_\omega(\widehat{\boldsymbol{\Omega}}) \equiv I^\alpha_\omega(\widehat{\boldsymbol{\Omega}}, \boldsymbol{r}_{\rm sc})$ (where the $\boldsymbol{r}_{\rm sc}$ dependence highlights the dependence on the scattering location) of mode α and frequency ω is given by (in the NS frame)

$$(\hat{\boldsymbol{\Omega}} \cdot \boldsymbol{\nabla})I^\alpha_\omega(\hat{\boldsymbol{\Omega}}) = -\int d\beta_e\, n_e f_1(\beta_e)(1-\beta_e\mu)\sigma_\alpha(\hat{\boldsymbol{\Omega}}, \beta_e)\, I^\alpha_\omega(\hat{\boldsymbol{\Omega}})$$
$$+\sum_\beta \int d\beta_e \oint {\rm d}\Omega'\, n_e f_1(\beta_e)(1-\beta_e\mu')\,\eta^{-3} I^\beta_{\eta\omega}(\hat{\boldsymbol{\Omega}}')$$
$$\times \left(\frac{d\sigma}{d\Omega'}\right)^{(\rm NS)}_{\beta\hat{\boldsymbol{\Omega}}'\rightarrow\alpha\hat{\boldsymbol{\Omega}}}, \tag{18}$$

where $\mu' = \widehat{\boldsymbol{\Omega}}' \cdot \widehat{\boldsymbol{B}}$ is measured in the NS frame, $\eta = \eta(\beta_e, \widehat{\boldsymbol{B}}, \widehat{\boldsymbol{\Omega}}', \widehat{\boldsymbol{\Omega}}) = (1-\beta_e^2)/[(1-\beta_e\cos\theta_1)(1+\beta_e\cos\theta'_2)]$ is the Doppler-shift factor linking the incident ($\eta\omega$) and scattered (ω) frequencies, $\cos\theta_1 = \widehat{\boldsymbol{\Omega}}' \cdot \widehat{\boldsymbol{B}}$ and $\cos\theta'_2 = \widehat{\boldsymbol{\Omega}} \cdot \widehat{\boldsymbol{B}}$ are calculated in the NS frame and ERF, respectively, and the η^{-3} factor represents the Doppler intensity rescaling through scattering (i.e., Comptonization). Note also that the factors $(1-\beta_e\mu)$ and $(1-\beta_e\mu'_0)$ in Equation (18) reflect the encounter rate due to relative motion.

We denote the position vector of a small emission patch on the NS surface by $R\widehat{\boldsymbol{\Omega}}_s$ and the position vector of the scattering location by $\boldsymbol{r}_{\rm sc} = r_{\rm sc}\widehat{\boldsymbol{\Omega}}_0$. Then the incoming direction vector can be calculated as $\widehat{\boldsymbol{\Omega}}' = (r_{\rm sc}\widehat{\boldsymbol{\Omega}}_0 - R\widehat{\boldsymbol{\Omega}}_s)/|r_{\rm sc}\widehat{\boldsymbol{\Omega}}_0 - R\widehat{\boldsymbol{\Omega}}_s|$. The first term on the right-hand side of Equation (18) represents the attenuation of the incoming intensity along $\widehat{\boldsymbol{\Omega}}$, and the second term represents the scattered-in contribution. For nonrelativistic electrons ($\beta_e \ll 1$), given the narrowness of the resonance, the scattering occurs on a thin surface. For single scattering, Equation (18) can be solved to yield the relationship between the incoming (before RCS) intensity $I^\alpha_{\omega,\rm in}$

[4] An exception occurs when the photon encounters the quasi-tangential (QT) point, where the photon momentum is nearly aligned with the local magnetic field. The X-ray polarization may change significantly when the photon passes through the QT region (C. Wang & D. Lai 2009). Since only a small fraction of the NS surface radiation is affected by the QT propagation, we shall neglect this effect in this paper.

and the outgoing (after RCS) intensity $I^{\alpha}_{\omega,\rm out}$ as

$$I^{\alpha}_{\omega,\rm out}(\widehat{\boldsymbol{\Omega}}) = I^{\alpha}_{\omega,\rm out}{}^{(\rm I)}(\widehat{\boldsymbol{\Omega}}) + I^{\alpha}_{\omega,\rm out}{}^{(\rm II)}(\widehat{\boldsymbol{\Omega}}). \tag{19}$$

The first term is the attenuated incoming intensity that propagates along $\widehat{\boldsymbol{\Omega}}$:

$$I^{\alpha}_{\omega,\rm out}{}^{(\rm I)}(\widehat{\boldsymbol{\Omega}}) = I^{\alpha}_{\omega,\rm in}(\widehat{\boldsymbol{\Omega}})\exp(-\tau_{\alpha}), \tag{20}$$

where $\tau_{\alpha}(\widehat{\boldsymbol{\Omega}}) = \int ds\, d\beta_e\, n_e f_1(\beta_e)(1-\beta_e\mu)\sigma_{\alpha}$ is the RCS optical depth for mode α. The second term in Equation (19) represents the scattered-in part, which is given by

$$I^{\alpha}_{\omega,\rm out}{}^{(\rm II)}(\hat{\boldsymbol{\Omega}}) = \sum_{\beta}\int ds\, d\beta_e \oint d\Omega'\, n_e f_1(\beta_e) \times (1-\beta_e\mu')\,\eta^{-3} I^{\beta}_{\eta\omega,\rm in}(\hat{\boldsymbol{\Omega}}')\left(\frac{d\sigma}{d\Omega'}\right)^{(\rm NS)}_{\beta\hat{\boldsymbol{\Omega}}'\to\alpha\hat{\boldsymbol{\Omega}}}. \tag{21}$$

In Equation (21), the integration over ds can be simplified using the delta function in the cross section, and the integration over $d\Omega'$ for a given scattering site $\boldsymbol{r}_{\rm sc}$ can be written as

$$\oint {\rm d}\Omega' = \oint \frac{d\boldsymbol{A}_s\cdot\hat{\boldsymbol{\Omega}}'}{|\boldsymbol{r}_{\rm sc}-R\hat{\boldsymbol{\Omega}}_s|^2} = \int \frac{dA_s(\hat{\boldsymbol{\Omega}}_s\cdot\hat{\boldsymbol{\Omega}}')}{|\boldsymbol{r}_{\rm sc}-R\hat{\boldsymbol{\Omega}}_s|^2}\Theta(\hat{\boldsymbol{\Omega}}_s\cdot\hat{\boldsymbol{\Omega}}'), \tag{22}$$

where $d\boldsymbol{A}_s = \hat{\boldsymbol{\Omega}}_s dA_s$ is the surface-element vector of the emission spot on the NS surface (see Figure 1). Note that we have included a Heaviside function $\Theta(\widehat{\boldsymbol{\Omega}}_s\cdot\widehat{\boldsymbol{\Omega}}')$ to ensure that only photons emitted outward from the surface are accounted for. Substituting the expression of n_e and $(d\sigma/d\Omega')$ with the simplifications above, Equation (21) can be rewritten as

$$I^{\alpha}_{\omega,\rm out}{}^{(\rm II)}(\hat{\boldsymbol{\Omega}}) = \int \frac{dA_s(\hat{\boldsymbol{\Omega}}_s\cdot\hat{\boldsymbol{\Omega}}')\Theta(\hat{\boldsymbol{\Omega}}_s\cdot\hat{\boldsymbol{\Omega}}')}{4\pi\,|\boldsymbol{r}_{\rm sc}-R\hat{\boldsymbol{\Omega}}_s|^2} \times \sum_{\beta}\int {\rm d}\beta_e \quad f_1(\beta_e)K_{\beta\alpha}(\hat{\boldsymbol{\Omega}}'\to\hat{\boldsymbol{\Omega}}) \times (1-\beta_e\mu')\eta^{-3} I^{\beta}_{\eta\omega,\rm in}(\hat{\boldsymbol{\Omega}}')\frac{\xi_{\tau}}{\beta_0}\frac{B}{r_{\rm sc}} \times \left|\frac{\partial}{\partial s}\left[\frac{B}{\gamma_e(1-\beta_e\mu)}\right]\right|^{-1}, \tag{23}$$

where $K_{\beta\alpha}(\widehat{\boldsymbol{\Omega}}'\to\widehat{\boldsymbol{\Omega}})$ is given by Equation (14). Equations (19), (20), and (23) give the total outgoing intensity after a single scattering event. We expect the single-scattering approximation to be valid when $\tau_{\alpha}\lesssim 1$. We may still apply it for the $\tau_{\alpha}\gtrsim 1$ cases to show qualitative physical trends. Higher-order solutions can be obtained perturbatively, analogous to those in M. Lyutikov & F. P. Gavriil (2006), where an analytical solution for the one-dimensional case is given.

After RCS, an outgoing photon continues to propagate in the magnetosphere, with its polarization state evolving adiabatically, i.e., $I^{\alpha}_{\omega}(\widehat{\boldsymbol{\Omega}})=$constant. Eventually, the photon reaches the "polarization-limiting radius" $r_{\rm pl}$, beyond which the polarization state is frozen. The value of $r_{\rm pl}$ depends on the photon energy E, the dipole field strength $B_{\rm p}$, and the magnetar rotation rate, and generally $r_{\rm pl}/R\gtrsim 100$ (M. van Adelsberg & D. Lai 2006). Since $r_{\rm pl}\gg r_{\rm sc}$, regardless of the structure and shape of the RCS layer, the radiation emerging from the layer with mode intensities I^{α}_{ω} evolves adiabatically in the magnetosphere such that the radiation at $r > r_{\rm pl}$ consists of approximately the same X-mode and O-mode intensities as those emerging from the RCS layer, with a small mixture of circular polarization generated around $r_{\rm pl}$. See D. Lai (2023) for more details.

2.4. Computational Workflow

In our magnetar radiation model, different physical processes are cleanly separated by their characteristic radial scales: emission near the surface, propagation and scattering through the magnetosphere, and polarization evolution from the scattering layer to the polarization-limiting radius. Our calculations are divided into three steps:

1. *Emission near the surface under vacuum resonance*. Near the NS surface and atmosphere, the dominant process is polarized X-ray emission influenced by QED vacuum resonance. This region has been extensively studied (e.g., M. van Adelsberg & D. Lai 2006; D. Lai 2023; R. M. E. Kelly et al. 2024), allowing us to directly adopt previously computed emissivity profiles as input. In this work, we use a simplified prescription for the surface emission, as described in Section 2.2.1.
2. *Adiabatic propagation from the surface and RCS*. As discussed, resonant scattering occurs at $\sim 10R$. After emission from the NS surface, the photon polarization evolves adiabatically following the magnetic field and propagates solely under gravitational effects until scattering. Light bending is handled using the method of A. M. Beloborodov (2002). Specifically, the light-bending mapping between local emission angle α and asymptotic angle ψ corresponds to a mapping between $(\widehat{\boldsymbol{\Omega}}_s\cdot\widehat{\boldsymbol{\Omega}}')$ at the RCS layer and that at the NS surface, which is used in Equation (23). Note that the dot product in Equation (23) represents the cosine of the asymptotic angle after light bending. We shall see later that this mapping is also applied in calculating the flux contribution of $I^{\alpha}_{\omega,\rm out}{}^{(\rm I)}$, where a similar dot product also appears.
3. *Continuing adiabatic propagation to large distances where polarization freezes*. After scattering, the polarization continues to evolve adiabatically until the photon reaches the polarization-limiting radius ($\gg r_{\rm sc}$), where the polarization freezes and aligns coherently, yielding the observable net polarization. The observed flux is computed by integrating the contributions from all locations on the scattering surface. Specifically, we integrate the mode-resolved intensity $I^{\alpha}_{\rm out,\omega}(\widehat{\boldsymbol{\Omega}}, \boldsymbol{r}_{\rm sc}(\widehat{\boldsymbol{\Omega}}_0))$ over the outer surface of the scattering layer.

Figure 1 illustrates the geometric framework for our calculations of the magnetar X-ray radiation. Panel (a) defines the geometry: We set up a Cartesian coordinate system with the magnetic axis along the z-axis and the observer's line of sight (LOS) in the y–z plane at an angle θ from the z-axis. The radiation source (position vector $R\widehat{\Omega}_s$) is located at (θ_s, ϕ_s) on the NS surface. The location on the scattering surface is denoted by $\boldsymbol{r}_{\rm sc} = r_{\rm sc}\widehat{\boldsymbol{\Omega}}_0$. The direction of the incoming photon is $\widehat{\boldsymbol{\Omega}}'$, and that of the scattered photon is $\widehat{\boldsymbol{\Omega}}$. Two photon populations contribute to the observed flux, as discussed in Section 2.3.2 and illustrated in Figure 1:

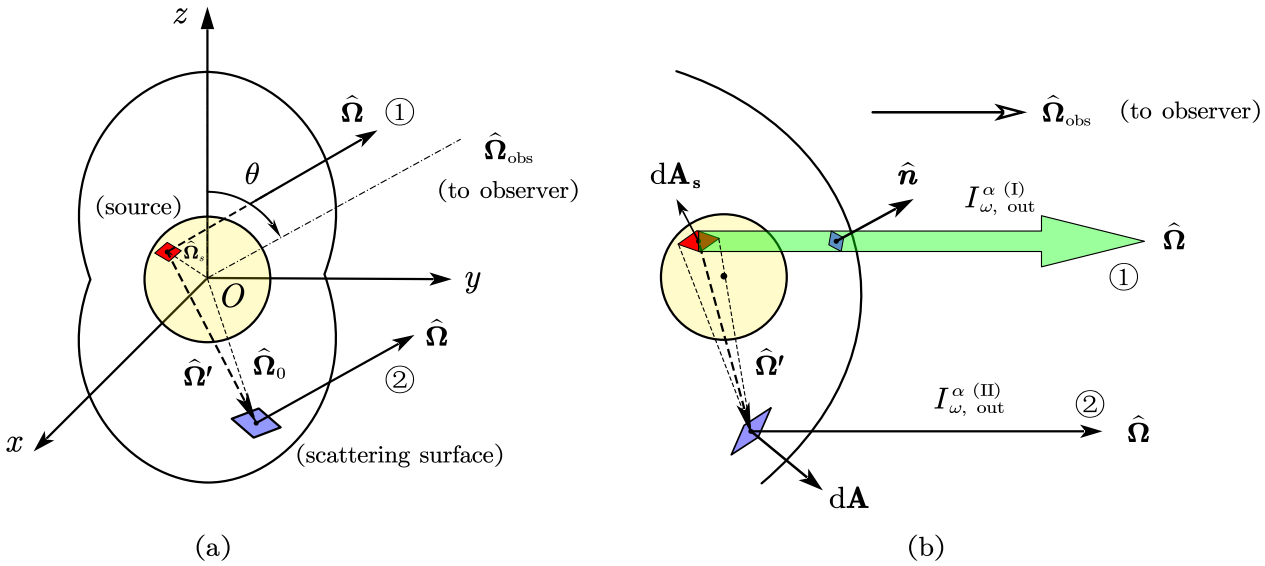


Figure 1. Schematic illustration of radiative transfer with resonant Compton scattering (RCS) and the computation of the observed polarization flux. The surface element of the NS is $d\boldsymbol{A}_s = \hat{\boldsymbol{\Omega}}_s dA_s$. The scattering surface element (located at $\boldsymbol{r}_{\rm sc} = r_{\rm sc}\hat{\boldsymbol{\Omega}}_{\rm sc}$) is denoted by $d\boldsymbol{A}$. The scattered radiation propagates along the direction $\hat{\boldsymbol{\Omega}} \simeq \hat{\boldsymbol{\Omega}}_{\rm obs}$, and the incident radiation (before scattering) propagates along $\hat{\boldsymbol{\Omega}}' = (r_{\rm sc}\hat{\boldsymbol{\Omega}}_0 - R\hat{\boldsymbol{\Omega}}_s)/|r_{\rm sc}\hat{\boldsymbol{\Omega}}_0 - R\hat{\boldsymbol{\Omega}}_s|$. The magnetic axis of the magnetar is along the z-axis, and $\hat{\boldsymbol{\Omega}}_{\rm obs}$ is in the y–z plane, with the angle between $\hat{\boldsymbol{\Omega}}_{\rm obs}$ and the z-axis being θ.

1. The attenuation term $I^{\alpha}_{\omega,\rm out}{}^{(\rm I)}(\hat{\boldsymbol{\Omega}})$: photons emitted directly toward the observer, attenuated by scattering.
2. The scattered-in term $I^{\alpha}_{\omega,\rm out}{}^{(\rm II)}(\hat{\boldsymbol{\Omega}})$: photons from all other directions that are scattered into the LOS.

Panel (b) of Figure 1 shows how scattered rays reach the observer. Ignoring stellar surface occultation, the total polarized flux in mode α is given by

$$\mathcal{F}^{\alpha}_{\omega} = \mathcal{F}^{\alpha}_{\omega}{}^{(\rm I)} + \mathcal{F}^{\alpha}_{\omega}{}^{(\rm II)} = \int I^{\alpha}_{\omega,\rm out}(\hat{\boldsymbol{\Omega}})(\hat{\boldsymbol{\Omega}}\cdot\hat{\boldsymbol{\Omega}}_{\rm obs})d\Omega. \quad (24)$$

Since the rays reaching the observer from different locations on the scattering surface are nearly parallel, we have $(\hat{\boldsymbol{\Omega}}\cdot\hat{\boldsymbol{\Omega}}_{\rm obs}) = 1$. Substituting Equations (19), (20), and (23) into Equation (24), using $d\Omega = |d\boldsymbol{A}\cdot\hat{\boldsymbol{\Omega}}_{\rm obs}|/D^2$, with $d\boldsymbol{A}$ being the area-element vector of the scattering surface and D being the distance to the observer, we obtain

$$\begin{aligned}
\mathcal{F}^{\alpha}_{\omega}{}^{(\rm I)} &= \frac{A_s}{D^2}(\hat{\boldsymbol{\Omega}}_s\cdot\hat{\boldsymbol{\Omega}}_{\rm obs})I^{\alpha}_{\omega,\rm in}(\hat{\boldsymbol{\Omega}}_{\rm obs})\exp(-\tau_\alpha),\\
\mathcal{F}^{\alpha}_{\omega}{}^{(\rm II)} &= \frac{A_s}{4\pi D^2}\frac{\xi_\tau}{\beta_0}\sum_{\beta}\int d\beta_e \oint d\Omega_0 \quad f_{\rm I}(\beta_e)\eta^{-3}\\
&\quad\times I^{\beta}_{\eta\omega,\rm in}(\hat{\boldsymbol{\Omega}}')K_{\beta\alpha}(\hat{\boldsymbol{\Omega}}'\to\hat{\boldsymbol{\Omega}}_{\rm obs})\\
&\quad\times(1-\beta_e\mu')\frac{r_{\rm sc}^2|\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{\Omega}}_{\rm obs}|}{|\boldsymbol{r}_{\rm sc}-R\hat{\boldsymbol{\Omega}}_s|^2}(\hat{\boldsymbol{\Omega}}_s\cdot\hat{\boldsymbol{\Omega}}')\Theta(\hat{\boldsymbol{\Omega}}_s\cdot\hat{\boldsymbol{\Omega}}')\\
&\quad\times\frac{B}{r_{\rm sc}}\left|\frac{\partial}{\partial s}\left[\frac{B}{\gamma_e(1-\beta_e\mu)}\right]\right|^{-1},
\end{aligned} \quad (25)$$

where we have assumed a small emission surface area A_s on the NS surface (in general, an integration over the whole NS surface can be performed). In Equation (25), we have used $d\boldsymbol{A} = \hat{\boldsymbol{n}}r_{\rm sc}^2\,d\Omega_0$, where $\hat{\boldsymbol{n}}$ is the surface normal vector of the scattering surface. Note that the final form of the first expression in Equation (25) does not involve an integration over the scattering surface, since only photons emitted directly toward the observer without scattering contribute to this term. Also note that, to simplify notation, the dot products $(\hat{\boldsymbol{\Omega}}_s\cdot\hat{\boldsymbol{\Omega}}_{\rm obs})$ and $(\hat{\boldsymbol{\Omega}}_s\cdot\hat{\boldsymbol{\Omega}}')$ in Equation (25) do not include GR effects. As mentioned before, when substituting the incoming intensity $I^{\alpha}_{\omega,\rm in}$, we must account for GR light bending from the NS surface to the scattering layer. In this context, the dot products $(\hat{\boldsymbol{\Omega}}_s\cdot\hat{\boldsymbol{\Omega}}_{\rm obs})$ and $(\hat{\boldsymbol{\Omega}}_s\cdot\hat{\boldsymbol{\Omega}}')$ represent the effective projection of the emission area onto the incoming direction $\hat{\boldsymbol{\Omega}}'$ after light bending. Similarly, by mapping $(\hat{\boldsymbol{\Omega}}_s\cdot\hat{\boldsymbol{\Omega}}_{\rm obs})$ back to the NS surface using the method of A. M. Beloborodov (2002), we can incorporate the GR light-bending effect in Equation (25).

After we obtain the fluxes $\mathcal{F}^{\alpha}_{\omega}$ of each mode, the linear polarization degree can be calculated as

$$P_{\rm L}(\omega) = \frac{\mathcal{F}^{2}_{\omega} - \mathcal{F}^{1}_{\omega}}{\mathcal{F}^{2}_{\omega} + \mathcal{F}^{1}_{\omega}}. \quad (26)$$

3. Result of the Simplified Fiducial Model

To identify how each ingredient in our semianalytic treatment shapes the observables, we begin with a simplified model for preliminary analysis. Throughout this section, we adopt a pure dipole field with a cold-plasma approximation, ignoring relativistic effects. The electron number density follows Equation (2), with the parameter $\zeta \equiv \xi_\tau/\beta_0$ controlling the electron density (see Equation (2)) and hence the effective RCS strength (see Equation (16)), independently of the field geometry. Under these assumptions, the mode-resolved optical depths can be calculated as $\tau^{(\rm NR)}_{\alpha} = \int ds\, n_e \sigma^{(\rm NR)}_{\alpha}$ and the flux

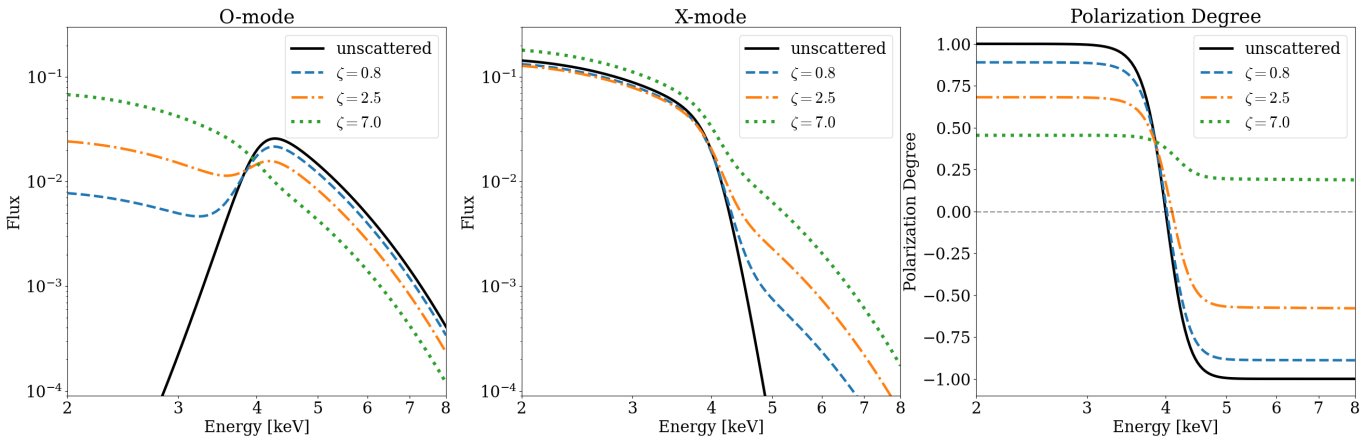


Figure 2. Flux and polarization spectra for different scattering strengths compared to the unscattered case in a simplified setup with $B_{\rm p} = 5 \times 10^{13}$ G, $kT = 0.6$ keV, $R = 10$ km, a hot spot at $(\theta_s, \phi_s) = (65°, 30°)$, viewing angle $\theta = 25°$, and varying $\zeta \equiv \xi_\tau/\beta_0 = 0.8$, 2.5, and 7.0 (see Equations (2) and (16)). Left and middle: the relative O-mode and X-mode fluxes. Right: the corresponding linear polarization degree $P_L(\omega)$ as a function of energy.

can be simplified to

$$\mathcal{F}_\omega^{\alpha({\rm NR,I})} = \frac{A_s \hat{\boldsymbol{\Omega}}_s \cdot \hat{\boldsymbol{\Omega}}_{\rm obs}}{D^2} I^\alpha_{\omega,{\rm in}}(\hat{\boldsymbol{\Omega}}_{\rm obs}) \exp(-\tau_\alpha^{({\rm NR})}),$$

$$\mathcal{F}_\omega^{\alpha\ ({\rm NR,II})} = \frac{A_s \zeta}{4\pi D^2} \sum_\beta \oint d\Omega_0 \quad K_{\beta\alpha} I^\beta_{\omega,{\rm in}}(\hat{\boldsymbol{\Omega}}') \frac{B}{r_{\rm sc}} \left| \frac{dB}{ds} \right|^{-1} \times \frac{r_{\rm sc}^2 |\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{\Omega}}_{\rm obs}|}{|\boldsymbol{r}_{\rm sc} - R\hat{\boldsymbol{\Omega}}_s|^2} (\hat{\boldsymbol{\Omega}}_s \cdot \hat{\boldsymbol{\Omega}}') \Theta(\hat{\boldsymbol{\Omega}}_s \cdot \hat{\boldsymbol{\Omega}}'). \tag{27}$$

For our fiducial setup, we adopt magnetar parameters $B_{\rm p} = 5 \times 10^{13}$ G, $R = 10$ km, and $kT_s = 0.6$ keV, with the emission point located at $(\theta_s, \phi_s) = (65°, 30°)$ and the observer's viewing angle set to $\theta = 25°$. The magnitude of $B_{\rm p}$ is consistent with the $B \lesssim B_{\rm QV}$ regime in polarized surface emission. We compute the polarized fluxes and linear polarization degree in the IXPE band (2–8 keV).

Figure 2 illustrates the impact of varying ζ on both the mode-resolved spectra and polarization of the observed radiation. For weaker scattering with small ζ (where the single-scattering approximation is valid), the initially dominant mode is attenuated, while the subdominant mode gains power through scattering, reducing the O- and X-mode spectral contrast. For stronger scattering with larger ζ, the total flux increases, since the scattered-in term scales directly with electron density. As the density becomes too high, our single-scattering approximation may overestimate the flux. In this regime, the spectral shapes of the O- and X-modes become similar.

The right panel of Figure 2 shows the corresponding linear polarization degree $P_{\rm L}(\omega)$ as a function of photon energy. As the RCS strength increases, $P_{\rm L}$ flattens, and it becomes positive for all photon energies in the case of $\zeta = 7$. Thus, the original mode-switching signature may disappear in strong-scattering cases. As we will see later, incorporating special-relativistic (SR) effects may erase the mode-switching signature more efficiently, even for cases with low optical depths. This phenomenon will be discussed in more detail in Sections 4 and 5.

Regarding the validity of our calculations, all the results above rely on the single-scattering approximation, i.e., a small resonant optical depth. Our numerical checks confirm that the estimated optical depth ($\sim$0.1–1) from Section 2 is valid. Specifically, for the geometry used above, we find $\tau \lesssim 1$ when $\zeta \lesssim 3.7$, supporting the accuracy of the single-scattering solution in this range. Notably, the disappearance of the mode switch in our calculations occurs at $\zeta \gtrsim 5.6$, where the single-scattering approximation becomes questionable.

For the high-magnetospheric-density cases, the scattered-in term is proportional to n_e and may therefore be overestimated in the first-order treatment. However, a fully self-consistent postscattering optical depth cannot be assigned within the simplified cold, nonrelativistic setup used in this section. In this baseline model the resonance is represented by a geometrically thin surface, and the velocity distribution is not resolved. However, any real current-conducting plasma has a finite velocity width such that a scattered photon can still resonate with electrons in the remaining part of the velocity distribution, and a photon scattered back toward the star can encounter additional resonant surfaces (another example of self-consistent treatment can be seen in K. Hu et al. 2026). A physical value of $\tau_{\alpha,{\rm esc}}$ therefore requires a genuine multiple-scattering calculation with a specified velocity distribution, scattering geometry, and photon trajectory.[5]

Here we use two limiting prescriptions only to place a limit on the possible impact of postscattering attenuation. The unweighted scattered-in term corresponds to our first-order baseline calculation, while the deliberately conservative prescription $\tau_{\alpha,{\rm esc}} = \tau_\alpha$ gives an upper-bound estimate of the suppression of the first-scattered component. In the latter case we define, with τ_α given by Equation (17),

$$\mathcal{F}_{\omega,limit}^{\alpha({\rm NR,I})} = \mathcal{F}_\omega^{\alpha({\rm NR,I})},$$
$$\mathcal{F}_{\omega,limit}^{\alpha({\rm NR,II})} = \mathcal{F}_\omega^{\alpha({\rm NR,II})} \exp(-\tau_\alpha). \tag{28}$$

Figure 3 shows the results for $\zeta = 7.0$, where $\tau_{\alpha,{\rm esc}} = 0$ and τ_α are adopted respectively as two limiting cases. As shown in the figure, after introducing a larger $\tau_{\alpha,{\rm esc}}$, the flux is substantially suppressed and therefore does not violate energy conservation. However, because the scattered-in contribution is also significantly weakened, the behavior originally

[5] The discussion on the contribution from higher-order terms in the following paragraphs aims at clarifying the reliability of the result that strong RCS may erase the mode-switching feature in the nonrelativistic regime; we do not include the effect of $\tau_{\rm esc}$ in Section 4.

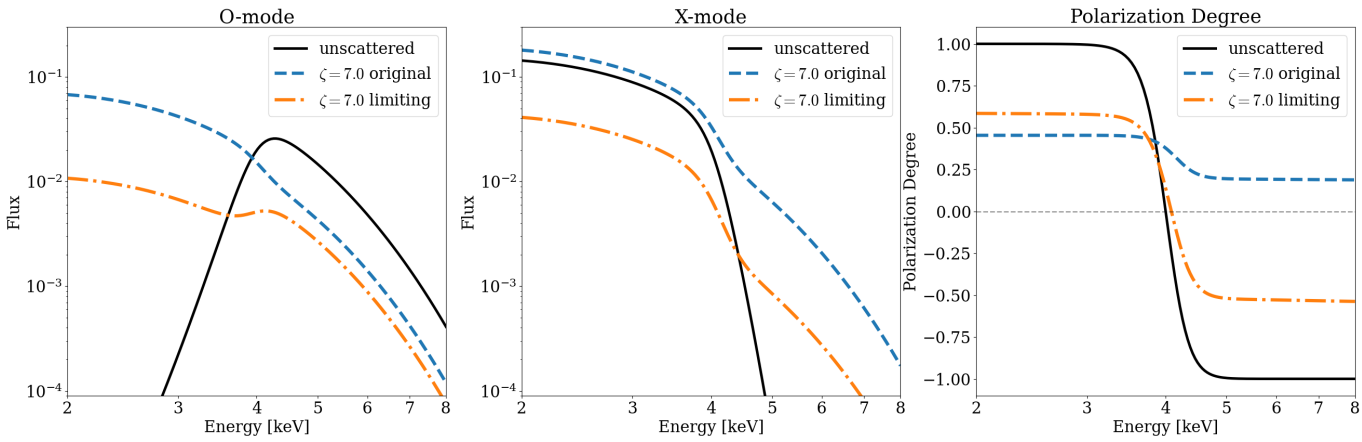


Figure 3. The relative O-mode and X-mode fluxes and polarization spectra for two first-order prescriptions (see Equation (28)) for the scattered-in component, compared with the unscattered case in the simplified model setup. The blue curves show the unweighted first-scattered baseline, while the orange curves apply the $\exp(-\tau_\alpha)$ attenuation factor to the scattered-in component. Other parameters are $B_p = 5 \times 10^{13}$ G, $kT = 0.6$ keV, $R = 10$ km, a hot spot at $(\theta_s, \phi_s) = (65°, 30°)$, viewing angle $\theta = 25°$, and $\zeta \equiv \xi_\tau/\beta_0 = 7.0$ (see Equations (2) and (16)).

indicated in the right panel, namely that RCS may erase the mode-switching feature, becomes less evident.

Note that in our full-model study in Section 4, once relativistic effects are included, RCS can still robustly produce the disappearance of the mode-switching behavior in the cases that generally satisfy $\tau \lesssim 0.4$. Thus, despite the single-scattering approximation, our calculation provides useful guidance on the effects of RCS on the magnetar soft X-ray polarization spectra.

4. Full Model: Parameter Study

In this section, we adopt the full-model setup, including both the thermal velocity distribution of magnetospheric electrons and (both special and general) relativistic effects. When the electron velocities are thermal about a mean drift, the resonance condition $\omega = \omega_D(\boldsymbol{r}, \mu, \beta_e)$ no longer selects a single geometric surface but broadens into a thin region centered around the drift-defined resonant surface, with a thickness set by the velocity distribution.

To include GR effects, we fix the compactness of the NS at

$$u \equiv \frac{2GM}{Rc^2} = 0.3, \tag{29}$$

and the other NS parameters are held consistent with the previous section. Since all photons experience nearly the same redshift from emission to scattering in our geometries, GR mainly introduces a common rescaling in the intensity and energy. Therefore, we interpret the surface temperature kT_s as the effective blackbody temperature of X-ray emission just before scattering, the same as in Section 3.

Under the current model, the scattering is controlled by the following parameter set:

$$\{\xi_\tau, \beta_0, kT_e, kT_s, \theta, M\}. \tag{30}$$

In the survey below, we fix kT_s, $E_{\rm ad}$, and M, and vary $(\xi_\tau, \beta_0, kT_e, \theta)$, while all other parameters are the same as in Section 3.

4.1. Optical-depth Dependence

Before analyzing the spectra and polarization, it is useful to delineate under what conditions the single-scattering solution

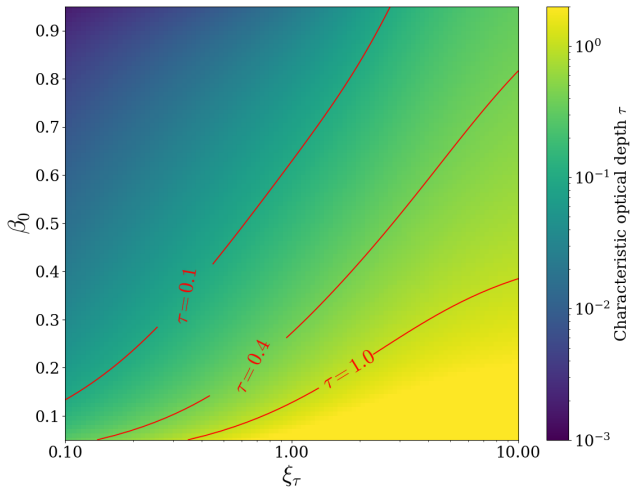


Figure 4 Heat map of the characteristic resonant optical depth in the full model. The parameters are as follows: $B_p = 5 \times 10^{13}$ G, $R = 10$ km, $kT_e = 10$ keV, $kT_s = 0.6$ keV, a hot spot at $(\theta_s, \phi_s) = (53°, 37°)$, and viewing angle $\theta = 35°$. Contours for $\tau_0 = 1.0$, 0.4, and 0.1 delineate the validity regime of the single-scattering/first-order approximation.

is valid. The resonant optical depth τ_α (for mode α) is primarily controlled by (ξ_τ, β_0), which set the electron density and relativistic beaming.

Figure 4 presents a heat map of the characteristic optical depth τ_0 (see Equation (16)), calculated at $kT_e = 10$ keV and $\theta = 35°$, as functions of ξ_τ and β_0. The $\tau_0 = 1$ contour marks the single-scattering boundary; the $\tau_0 = 0.4$ contour corresponds to $\lesssim$10% error for a first-order expansion of $\exp(-\tau)$. As expected, larger ξ_τ and smaller β_0 yield higher τ_α due to the increased particle density. Physically, the optical-depth map delineates the regime of applicability for the results in this section: For parameter combinations with $\tau \lesssim 1$, the single-scattering approximation may be regarded as approximately valid; for parameter combinations with $\tau \lesssim 0.4$, the single-scattering solution is expected to be reasonably reliable.

In addition, the optical-depth map also provides a useful reference for interpreting the results in the subsequent parameter study: For instance, if a certain behavior is observed in the regime $\tau \lesssim 0.4$, it can be confidently attributed to the

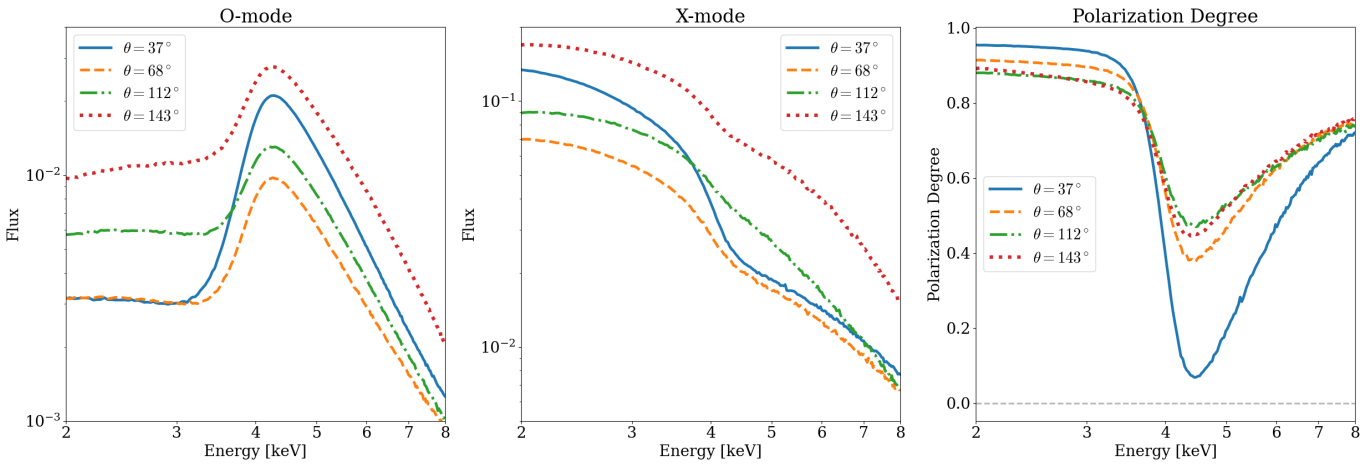


Figure 5. Flux and polarization spectra for different observer viewing angles in the full model. The parameters are as follows: $B_{\rm p} = 5 \times 10^{13}$ G, $R = 10$ km, $kT_e = 10$ keV, $kT_s = 0.6$ keV, $\xi_\tau = 0.2$, $\beta_0 = 0.5$, and a hot spot at $(\theta_s, \phi_s) = (90°, 37°)$. Results are shown for viewing angles $\theta = 37°$, 68°, 112°, and 143°.

RCS effect; if it is observed in the regime $\tau \gtrsim 1$, it should not be regarded as a strict quantitative prediction of RCS.

Using the twist parameterization defined in Equation (1), the total pole-to-pole twist angle is approximately

$$\Delta\phi = \int_0^\pi \frac{B_\phi}{B_\theta \sin\theta}\, d\theta \sim \pi\xi_\tau, \tag{31}$$

so the ranges adopted in previous Monte Carlo studies (e.g., L. Nobili et al. 2008a; R. Fernández & S. W. Davis 2011) fall largely within the $\tau_0 \lesssim 1$ regime in Figure 4.

4.2. Spectra and Polarization

We survey four key parameters: the viewing angle θ, the magnetic twist ξ_τ, the mean drift speed of electrons β_0, and the plasma temperature kT_e. The observed flux is computed as a multidimensional integral over the emission direction, scattering geometry, and electron velocity. Given the smoothness of the integration kernel, we evaluate the integral via Monte Carlo quadrature with 10^6 samples per configuration. This choice introduces small statistical fluctuations, or "noise," in the results, but enables significantly improved computational speed and numerical stability, resulting in a relative numerical uncertainty below 0.5%.

Viewing geometry. Figure 5 presents the flux and polarization spectra for various viewing angles. For this plot, we consider a single hot spot located at the equator $(\theta_s, \phi_s) = (90°, 37°)$ to highlight the north–south asymmetries (see L. Nobili et al. 2008a). Notably, pairs of LOS angles symmetric about the equator (e.g., 37° and 143°) yield markedly different flux and polarization spectra, demonstrating the impact of Doppler shifts caused by large-scale net currents flowing from magnetic north to south. Importantly, the polarization curves retain the qualitative behavior previously seen in Section 3: RCS may suppress or erase the intrinsic mode-conversion feature that would otherwise appear in the emission. For instance, the curve corresponding to $\theta = 37°$ shows sign flips in polarization degree, whereas for other angles $P_{\rm L}$ remains positive across the energy band. We verify that, in this parameter setting, the optical depth $\tau_\alpha(\theta)$ remains well below unity (typically $\lesssim$0.15), indicating that even single-scattering events can significantly modify the polarization signature, despite the overall low scattering probability.

Magnetic twist. Figure 6 explores how varying the magnetic twist ξ_τ affects the resulting flux and polarization spectra. The twist primarily modulates the plasma density and hence the scattering strength. As ξ_τ increases, the flux spectra show progressively larger deviations from the unscattered case, exhibiting the "spectral smoothing" noted in Section 3. However, relativistic effects become more prominent in the polarization spectra, especially at higher photon energies. Compared to the nonrelativistic case, SR effects cause the polarization degree to rise again at higher energies. In some cases, the polarization curve crosses zero twice within the IXPE band, indicating the possibility of multiple 90° PA swings (this is also seen in a pure QED scenario; see D. Lai 2023; R. M. E. Kelly et al. 2024). For sufficiently large ξ_τ, polarization remains strictly positive and mode switching disappears.

Electron drift velocity. Figure 7 shows the impact of changing the average drift velocity β_0 on the flux and polarization spectra. Like ξ_τ, the parameter β_0 influences both the plasma density and the strength of SR beaming. While its effects on the flux spectrum resemble those of increasing ξ_τ, the behavior of the polarization curves offers further diagnostic insight. Decreasing β_0 increases the particle density and weakens SR beaming, suppressing $P_{\rm L}$ at low energies and flattening its high-energy tail, which was initially raised by SR; this is consistent with the nonrelativistic trend discussed in Section 3. However, in terms of the first zero-crossing energy—the location where the initial 90° PA swing occurs—the effect of varying β_0 is subtler than that of varying ξ_τ. The transition slopes remain similar, indicating that β_0 mainly sets the vertical offset (baseline) of $P_{\rm L}$, while it plays a secondary role in determining the slope that shapes the crossing location.[6]

Together, these two main control parameters, ξ_τ and β_0, determine how RCS reshapes polarization spectra, mainly through changes in the magnetospheric density and SR effects. Two principal effects emerge: (i) a reduction in both the absolute value and spectral variation of the polarization

[6] A dynamic analogy may be helpful: Imagine the polarization curve as a catenary suspended between a high left platform and a vertical wall on the right. Increasing the relativistic effect raises the right anchor, steepening the tail of the curve while leaving its middle largely unchanged.

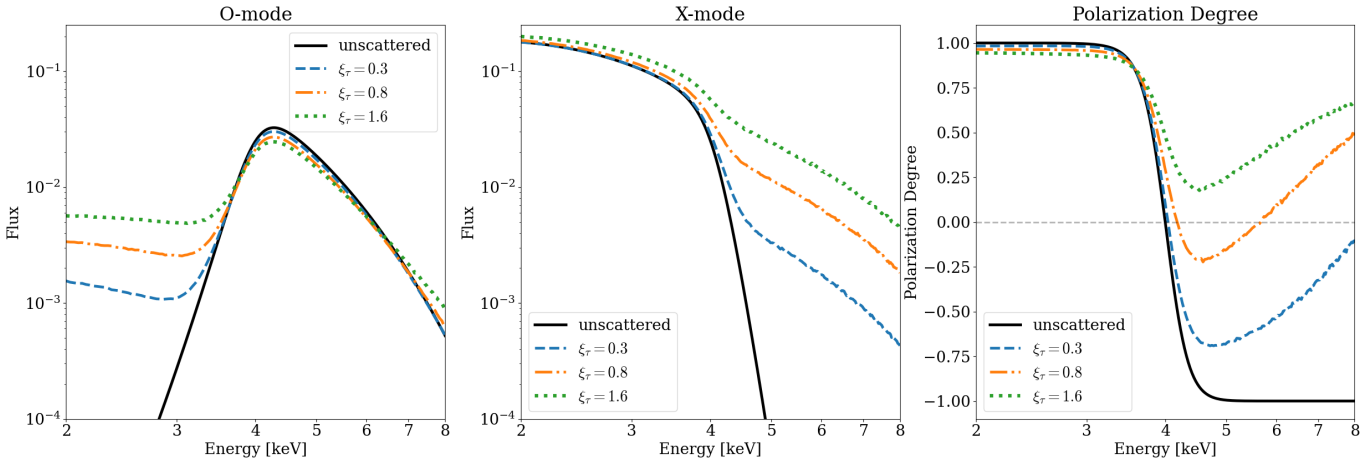


Figure 6. Flux and polarization spectra for different magnetospheric twist parameters in the full model. The parameters are as follows: $B_p = 5 \times 10^{13}$ G, $R = 10$ km, $kT_e = 10$ keV, $kT_s = 0.6$ keV, $\theta = 35°$, $\beta_0 = 0.5$, and a hot spot at $(\theta_s, \phi_s) = (53°, 37°)$. Results are shown for magnetic twists $\xi_\tau = 0.3$, 0.8, and 1.6, and are compared with the unscattered case.

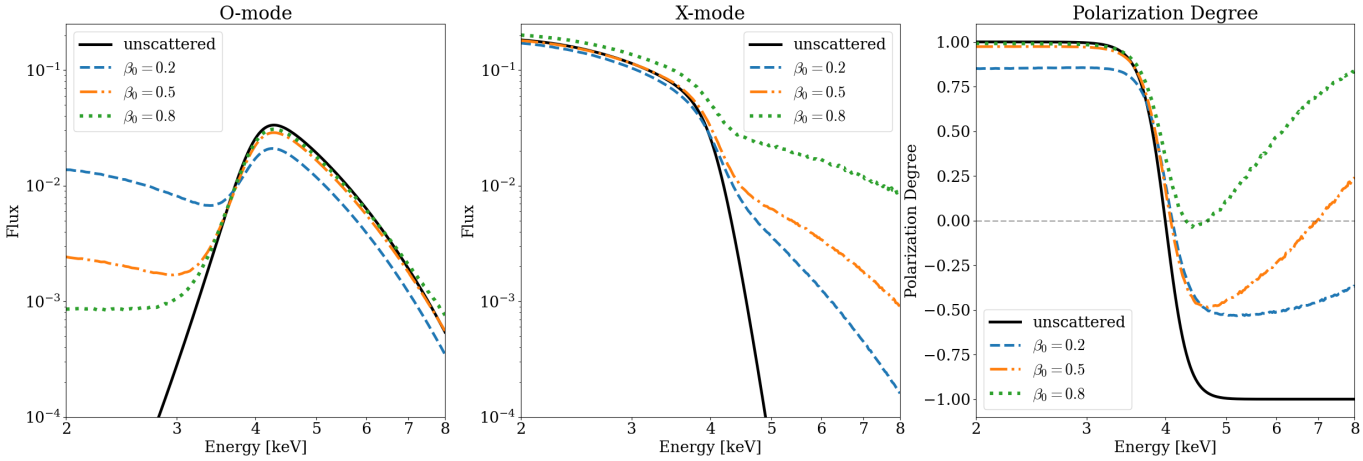


Figure 7. Flux and polarization spectra for different mean electron drift velocities in the full model. The parameters are as follows: $B_p = 5 \times 10^{13}$ G, $R = 10$ km, $kT_e = 10$ keV, $kT_s = 0.6$ keV, $\theta = 35°$, $\xi_\tau = 0.5$, and a hot spot at $(\theta_s, \phi_s) = (53°, 37°)$. Results are shown for $\beta_0 = 0.2$, 0.5, and 0.8, and are compared with the unscattered case.

degree, and (ii) a high-energy rise in P_L driven by SR mode redistribution. Even in the single-scattering regime, RCS may erase the mode-switch feature that is originally present in the emission. To understand this trend, consider the polarized RCS cross sections in the ERF (see Equation (10)):

$$\sigma_{1-1} = \frac{1}{3}\sigma_{1-2} = \frac{\pi^2 e^2}{2m_e c}\delta(\omega - \omega_B)(\boldsymbol{\Omega}' \cdot \hat{\boldsymbol{B}})^2,$$
$$\sigma_{2-2} = 3\sigma_{2-1} = \frac{3\pi^2 e^2}{2m_e c}\delta(\omega - \omega_B). \quad (32)$$

These expressions represent the relative probabilities for photons to scatter between specific initial and final polarization states. Notably, regardless of the initial state, scattering tends to populate the X-mode (polarization index $\alpha = 2$) more efficiently. Since the original emission is X-mode dominated at low energies, RCS has a limited effect there, whereas at higher energies (O-mode dominated), RCS transfers power to the X-mode, producing the characteristic high-energy rise in P_L. The electron density mainly rescales P_L, while the SR effects modulate its slope, particularly in the high-energy tail. Together, they reshape the polarization distribution and influence the energy at which mode switches occur.

Thermal broadening. Finally, Figure 8 examines the role of the plasma temperature kT_e in shaping the flux and polarization spectra. For visibility, the flux panels are vertically rescaled. Compared to the single-velocity case, thermal motion induces only modest changes. Unlike β_0, which parameterizes the bulk drift, increasing kT_e only broadens the velocity distribution around β_0. This tends to average out the variation in Doppler effects over electron speeds, resulting in behavior that closely resembles the single-velocity case at the same β_0. This is consistent with previous findings by L. Nobili et al. (2008a).

Power-law-like spectral component. Previous Monte Carlo simulations of magnetar spectra show that repeated RCS can convert seed thermal photons into a nonthermal component (e.g., L. Nobili et al. 2008a, 2008b). However, our single-scattering calculation is not originally designed to produce such a fully developed continuum, but it does capture the onset of spectral hardening by a single resonant scattering event. To

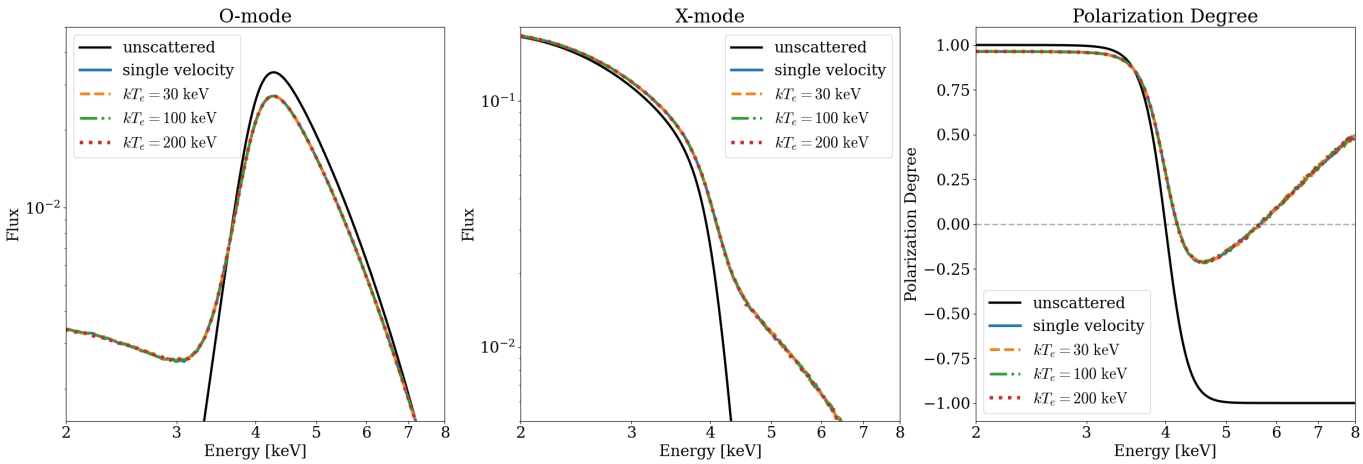


Figure 8. Flux and polarization spectra for different magnetospheric plasma temperatures in the full model. The parameters are as follows: $B_p = 5 \times 10^{13}$ G, $R = 10$ km, $kT_s = 0.6$ keV, $\theta = 35°$, $\xi_\tau = 0.8$, $\beta_0 = 0.5$, and a hot spot at $(\theta_s, \phi_s) = (53°, 37°)$. Results are shown for $kT_e = 30$, 100, and 200 keV, and are compared with the unscattered and single-velocity cases.

quantify this effect, we fitted the calculated photon spectrum in the IXPE band with a blackbody-plus-power-law model:

$$\dot{N}(E) = \dot{N}_{\rm BB}(E) + \dot{N}_{\rm PL}(E) = A_{\rm BB}\frac{E^2}{\exp(E/kT_s) - 1} + A_{\rm PL}E^{-\Gamma_{\rm eff}}, \quad (33)$$

where $\dot{N}$ represents the total photon number flux, and $\dot{N}_{\rm BB}$ and $\dot{N}_{\rm PL}$ are the contributions from the blackbody and power-law components, respectively. For the representative $\xi_\tau = 0.8$ case in Figure 6, the XSPEC fit gives $\Gamma_{\rm eff} \simeq 3.0$. The power-law-like residual contributes only about 18% of the integrated flux at $\sim$2 keV and becomes dominant mainly above $\sim$6 keV. A local estimate of the residual slope, $-d\ln\dot{N}_{\rm PL}/d\ln E$, varies strongly across the band from order unity to higher than 4, rather than forming a stable broadband power law. We therefore interpret this component as a local hard excess produced by the onset of Comptonization, not as a self-consistent nonthermal continuum. Such a continuum requires higher-order, iterative resonant inverse Compton scatterings. We also note that the blackbody-plus-power-law decomposition is phenomenological; for some other soft X-ray magnetar spectra, a two-temperature blackbody model can provide a comparably good fit (e.g., Z. Wadiasingh et al. 2026).

5. Summary and Discussion

In this work, we have developed a unified and analytically tractable framework to approximate how RCS in the magnetosphere and QED vacuum resonance in the atmosphere jointly shape the X-ray spectra and polarization observed from magnetars. Starting from polarized radiative transfer (Equation (18)) with scattering, we adopt a first-order approximation in optical depth (Equations (19), (20), and (23)), which effectively captures single-scattering events, to compute the observed polarization fluxes (Equations (24)–(26)). For a concrete baseline, we adopt an idealized initial surface-emission polarization pattern that mimics vacuum-resonant conversion—100% X-mode below 4 keV and 100% O-mode above, with a smooth transition in between (D. Lai 2023)—and then quantify how RCS in the magnetosphere reshapes the observable polarization signatures. Our key findings are as follows:

1. Increasing the plasma density in the magnetosphere leads to an overall suppression of both the absolute value and spectral variation of polarization degree $P_{\rm L}$. In other words, RCS acts to compress the polarization curve toward a horizontal line located *above* zero, diminishing the polarization contrast (see Figures 2 and 6).
2. Increasing the average electron drift velocity enhances SR effects, which tend to raise $P_{\rm L}$ in the high-energy tail and steepen its slope. In some cases, an extra mode switch (i.e., sign change in $P_{\rm L}$, implying a 90° PA swing) induced by RCS may occur (see Figure 7).
3. The energy of the *first* mode switch is only weakly shifted by RCS in the parameter ranges explored here; the zero-crossing location remains largely tied to the stellar surface physics set by vacuum resonance.
4. In the strong-RCS regime, the combined effects of increased plasma density and enhanced SR effects may entirely erase the signature of a mode switch within the observed bandpass (2–8 keV), even when QED-induced conversion is present in the initial emission. This is consistent with the mode-redistribution trend expected from the resonant cross sections (see Equation (32)).

However, these conclusions should be interpreted in light of several simplifying assumptions.

(i) Our magnetospheric configuration is *prescribed* rather than dynamically self-consistent. As discussed in Section 2, the field and current systems are parameterized and do not enforce full force-free equilibrium. In addition, the actual magnetosphere is likely to be partly twisted rather than globally twisted, and the prescribed thermal velocity distribution in the magnetosphere may not capture some intrinsic physical effects (e.g., R. Fernández & C. Thompson 2007, who considered a power-law distribution of relativistic momentum). Also, the electron density prescription applied in this work might underestimate the RCS optical depth of the magnetosphere. Indeed, recent kinetic modeling of magnetar coronae suggests that the pair multiplicities can be moderately high (C. Thompson 2008; A. M. Beloborodov 2013; C. Thompson & A. Kostenko 2020; J. Zhang & C. Thompson 2025), i.e., the total density

$n_{e^+} + n_{e^-} \gtrsim 30 n_e$. If so, the single-scattering approximation may be questionable. Nevertheless, we still expect that the main trends identified in this work, especially the trend of RCS erasing the mode-switching feature, will be qualitatively robust, since they are primarily driven by the relative scattering probabilities between modes (Equation (32)), which are independent of the overall density.

(ii) Our sample results consider emission from a single surface patch. Radiation from the whole NS surface can be easily included in our formalism and will likely average out some geometric contrasts. Also, a realistic configuration of the extended hot spot may introduce additional behavior in the polarization spectra, which is beyond the scope of this work.

(iii) Our transport solution is only of first order in optical depth. Fully capturing the energy-dependent behavior (e.g., the power-law high-energy tail and Comptonization) and energy/polarization redistribution at high RCS optical depth requires higher-order solutions via iterative scattering treatments (e.g., one-dimensional iterative scattering; see M. Lyutikov & F. P. Gavriil 2006).

(iv) Our adopted surface-polarization pattern is idealized; a more realistic treatment would require detailed modeling of the NS surface and atmosphere.

(v) Finally, we emphasize the restricted domain of validity of the imposed surface-polarization prescription. The atmospheric vacuum-resonance origin of the seed mode switch is self-consistent only when the local magnetic field in the emitting region satisfies $B_{\rm emission} \lesssim B_{\rm QV}$, with $B_{\rm QV}$ given by Equation (7). For the fiducial parameters in our model, this corresponds to a $B_{\rm QV}$ of a few 10^{13} G across the IXPE band, with variations by factors of a few depending on composition and geometry. This is consistent with the observation of 4U 0142+61 (R. Taverna et al. 2022). If $B_{\rm em} \gtrsim B_{\rm QV}$, the surface emission is expected to remain X-mode dominated over the band, and the atmospheric vacuum resonance alone would not generate the intrinsic 90° PA swing assumed in our illustrative seed model. Our results should therefore be interpreted as applying either to low-local-field emitting regions, or to cases where a similar seed polarization pattern is produced by another surface-emission mechanism. The global magnetic-energy budget may still be dominated by hidden internal or toroidal fields, which are not modeled in the present calculation.

Despite these limitations, the semianalytic nature of our method offers a major computational advantage and physical transparency. It allows us to extract the main physical trends without the computational cost or complexity of full Monte Carlo simulations and reveals intuitive connections between model inputs and observable outcomes.

Looking ahead, two immediate extensions are natural. First, to better capture viewing-angle-dependent effects, full-surface integration should be implemented, which would remove projection biases from small-area sources and allow more realistic modeling of observation profiles. In our formalism, this only requires treating the coordinates of the surface-emission point as integration variables over solid angle. Second, including stellar rotation would allow phase-resolved predictions when the magnetic and spin axes are misaligned. In our framework, this is equivalent to a rotation of the LOS direction vector and tracking the evolution of the polarization signal with phase, enabling the calculation of phase-resolved light curves and phase-averaged polarization profiles. These developments, together with higher-order scattering, will help connect magnetar surface and magnetosphere models with observations. Upcoming X-ray polarization missions, such as eXTP (S. Zhang et al. 2016, 2019), will have greatly enhanced polarization sensitivity and an extended energy band, providing simultaneous spectral, timing, and polarization data and thereby enriching the magnetar polarization sample for further studies.

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